Lecture 04 - Exercises

These exercises accompany Lecture 04, "Probability Foundations and Random Variables." They practice the probability language needed to reason about uncertain business and finance outcomes: sample spaces, events, probability rules, conditional probability, independence, random variables, expected value, variance, and standard deviation.

The exercises are separated into two parts. The first part should be done by hand. This is important because manual work helps you understand the mechanics: what the sample space is, which event is being counted, which denominator is being used, and how probability-weighted averages are built. Only after that does it become useful and convenient to move to larger datasets and use Python.

The second part uses Python. These exercises apply the same ideas to a return dataset where manual counting would be inefficient. Python is used to create event indicators, estimate probabilities from relative frequencies, check conditional probabilities, compare observed joint events with an independence benchmark, and write careful interpretations.

Solutions are not included here. They will be provided separately in a notebook.

Part A - By-Hand Practice

Exercise 1 - Outcomes, Sample Space, and Events

A one-month business project has three possible demand states:

Demand state	Project profit
Strong	100
Normal	30
Weak	-70

1. Write the sample space for the demand state.
2. Define the event A: "the project makes a positive profit."
3. Define the event B: "the project loses money."
4. Are A and B mutually exclusive? Explain.
5. Are A and B collectively exhaustive? Explain.

Exercise 2 - Assigning Probabilities

A product launch has the following possible outcomes:

Outcome	Probability
Failure	0.30
Moderate success	0.50
High success	0.20

1. Check that the probabilities define a valid probability model.
2. Interpret the probability of "Failure" in words.
3. Define the event S: "the launch is successful," where success means moderate success or high success.
4. Compute $P(S)$.
5. Define S^c in words and compute $P(S^c)$.

Exercise 3 - Complement Rule

The probability that monthly sales are below target is 35%.

Let A be the event "monthly sales are below target."

1. Write $P(A)$.
2. Define the complement A^c in words.
3. Compute $P(A^c)$.
4. Interpret the result in one sentence.

Exercise 4 - Union and Intersection

Let:

$$A = \{\text{a customer churns this quarter}\}$$

and

$$B = \{\text{the customer had a service complaint this quarter}\}.$$

Suppose:

$$P(A) = 0.40, \qquad P(B) = 0.30, \qquad P(A \cap B) = 0.18.$$

1. Interpret $P(A)$, $P(B)$, and $P(A \cap B)$ in words.
2. Compute $P(A \cup B)$.
3. Interpret $P(A \cup B)$ in words.
4. Explain why the union formula subtracts $P(A \cap B)$.

Exercise 5 - Mutually Exclusive Events on a Return Line

Let R be a one-day return. Define:

$$A = \{R < -2\%\},$$

$$B = \{R > 3\%\},$$

$$C = \{R > 0\},$$

and

$$D = \{|R| < 3\%\}.$$

For each pair, say whether the events are mutually exclusive and explain briefly.

1. A and B.
2. A and C.
3. B and C.
4. A and D.
5. B and D.
6. C and D.

Exercise 6 - Conditional Probability

Let:

$$A = \{\text{a delivery arrives more than two days late}\}$$

and

$$B = \{\text{there is a supplier disruption during the same week}\}.$$

Suppose:

$$P(A \cap B) = 0.08, \qquad P(B) = 0.12.$$

1. Compute $P(A \mid B)$.
2. Interpret the result in words.
3. Explain why the denominator is $P(B)$, not $P(A)$.

Exercise 7 - Do Not Reverse the Conditioning

Let:

$$D = \{\text{a customer churns}\}$$

and

$$F = \{\text{the customer experienced a service failure}\}.$$

Suppose:

$$P(D \cap F) = 0.03, \qquad P(F) = 0.10, \qquad P(D) = 0.05.$$

1. Compute $P(D \mid F)$.
2. Compute $P(F \mid D)$.
3. Are these two probabilities equal?
4. Explain why they answer different questions.

Exercise 8 - Conditional Probability from a Two-Way Table

A portfolio contains 42 firms:

Region	Telecom	Banking	Total
Europe	5	12	17
US	18	7	25
Total	23	19	42

One firm is selected at random.

1. Compute $P(\text{Banking})$.
2. Compute $P(\text{US})$.
3. Compute $P(\text{Banking} \cap \text{US})$.

4. Compute $P(\text{Banking} \mid \text{US})$.
5. Compute $P(\text{US} \mid \text{Banking})$.
6. Explain why $P(\text{Banking} \mid \text{US})$ and $P(\text{US} \mid \text{Banking})$ are different.

Exercise 9 - Independence Check

Let:

$$A = \{\text{Product A has below-target weekly sales}\}$$

and

$$B = \{\text{Product B has below-target weekly sales}\}.$$

Suppose:

$$P(A) = 0.45, \qquad P(B) = 0.40, \qquad P(A \cap B) = 0.25.$$

1. Compute $P(A)P(B)$.
2. Compare $P(A)P(B)$ with $P(A \cap B)$.
3. Are A and B independent?
4. Interpret your answer in words.

Exercise 10 - Base Rates and False Positives

A risk flag is used to identify firms that may default within one year.

True outcome	Flagged high risk	Not flagged high risk	Total
Defaults	30	20	50
Does not default	190	760	950
Total	220	780	1000

Let:

$$D = \{\text{the firm defaults}\}$$

and

$$F = \{\text{the firm is flagged high risk}\}.$$

1. Compute $P(D)$.
2. Compute $P(F)$.
3. Compute $P(F \mid D)$.
4. Compute $P(D \mid F)$.
5. Explain why $P(F \mid D)$ and $P(D \mid F)$ are different.
6. Is the flag useless because $P(D \mid F)$ is much lower than 50%? Explain carefully.

Exercise 11 - Random Variable, Realization, Event, or Probability?

Classify each statement as referring mainly to a random variable, a realization, an event, or a probability.

Statement	Classification
Next month's sales before the month begins	
Yesterday's stock return was -1.4%	
The project loses money next quarter	
The probability of a customer churning is 4%	
The number of defaults in a loan portfolio next year	
The observed number of late deliveries last month was 12	
The stock return is below -2% tomorrow	
$P(R < 0) = 0.35$	

For two rows, explain your classification.

Exercise 12 - From Outcomes to Random Variables

Suppose we toss a fair coin three times. The sample space is:

$$\{HHH,\ HHT,\ HTH,\ HTT,\ THH,\ THT,\ TTH,\ TTT\}.$$

Each outcome has probability 1/8. Define:

$$X = 1 \text{ point for each Head and -1 point for each Tail}$$

and

$$Y = 10 \text{ points if all three tosses are equal, and 0 otherwise.}$$

1. Complete a table with the value of X and Y for each outcome.
2. Build the probability distribution of X.
3. Build the probability distribution of Y.
4. Explain why the same sample space can generate different random variables.

Exercise 13 - Valid Probability Distribution

A future demand state is modeled as follows:

Demand state	Probability
Strong	0.25
Normal	0.50
Weak	0.25

1. Check that this is a valid probability distribution.
2. What must be true for a table to be a valid probability distribution?
3. If another analyst wrote probabilities $0.25, 0.50, 0.35$, what would be wrong?

Exercise 14 - Expected Profit

A project has the following one-month profit distribution:

Demand state	Probability	Profit
Strong	0.30	140
Normal	0.50	40
Weak	0.20	-90

1. Check that the probabilities sum to 1.
2. Compute the expected profit.
3. Is the expected profit necessarily one of the possible realized profits?
4. Interpret the expected profit in words.

Exercise 15 - Variance and Standard Deviation

Use the profit distribution from Exercise 14.

1. Use your answer from Exercise 14 to identify $E[X]$.
2. Compute the deviation of each profit from $E[X]$.
3. Square each deviation.
4. Weight each squared deviation by its probability.
5. Compute $Var(X)$.
6. Compute $SD(X)$.
7. Interpret the standard deviation in words.

Exercise 16 - Same Expected Return, Different Risk

Two investments have the following return distributions:

Investment	Outcome	Probability	Return
A	Up	0.50	8%
A	Down	0.50	2%
B	Up	0.50	14%
B	Down	0.50	-4%

1. Compute the expected return of Investment A.
2. Compute the expected return of Investment B.
3. Compare the two expected returns.
4. Compute the standard deviation of each investment's return.
5. Which investment would you describe as riskier? Explain.

Exercise 17 - Careful or Too Strong?

For each statement, say whether it is careful or too strong. Rewrite if needed.

1. "The probability of customer churn is 4%, so the customer will not churn."
2. "The expected profit is a probability-weighted average of possible profits."
3. "The expected profit is 3, so the realized profit must be 3."
4. "The sample frequency of below-target sales was 30%, which may be used as an estimate of the probability of below-target sales if the sample is relevant."
5. "Because two products both had below-target sales last week, their sales outcomes are independent."
6. "The probability of a portfolio loss may be higher conditional on a market downturn."
7. "The expected return summarizes the center of the return distribution, but it does not fully describe risk."

8. "A higher standard deviation means that realized outcomes tend to be more dispersed around the expected value."

Part B - Python Applications

For the Python exercises, use `numpy`, `pandas`, and `matplotlib.pyplot` unless stated otherwise.

Use the file:

`data and notebooks/Lecture 04 market loss days.csv`

The file contains 180 trading days with columns `date`, `market_return`, and `stock_return`.

Exercise 18 - Python: Build Event Variables from Returns

1. Load the CSV file.
2. Convert `date` to a date column.
3. Create the following Boolean event variables:
   - `market_stress`: market return below -1%;
   - `stock_loss`: stock return below 0;
   - `large_stock_loss`: stock return below -2%;
   - `same_direction`: market and stock returns have the same sign.
4. Display the first rows of the dataset with the new event variables.
5. Compute the relative frequency of each event.
6. Explain why these relative frequencies are descriptive statistics before they are used as probability estimates.

Exercise 19 - Python: Probability Rules from Event Flags

Let:

$$A = \{\text{market stress}\}$$

and

$$B = \{\text{large stock loss}\}.$$

Using the event variables from Exercise 18:

1. Estimate $P(A)$.
2. Estimate $P(A^c)$.
3. Estimate $P(B)$.
4. Estimate $P(A \cap B)$.
5. Estimate $P(A \cup B)$.
6. Check that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
7. Interpret the union probability in words.

Exercise 20 - Python: Conditional Probability and Risk Lift

Using the same events A and B:

1. Estimate $P(B \mid A)$, the probability of a large stock loss on market-stress days.
2. Estimate $P(B \mid A^c)$, the probability of a large stock loss on non-stress days.
3. Compute the risk lift:

$$\frac{P(B \mid A)}{P(B \mid A^c)}.$$

4. Interpret the risk lift.
5. Explain why this is evidence of association in this sample, not necessarily a causal claim.

Exercise 21 - Python: Independence Benchmark

Using the same events A and B:

1. Count the observed number of days where both A and B occur.
2. Compute the expected number of joint days under independence:

$$nP(A)P(B).$$

3. Compare the observed count with the independence benchmark.
4. Does the sample suggest that market stress and large stock losses behave independently?
5. Explain why independence is stronger than simply saying that two events can occur together.

Exercise 22 - Python: Build a Discrete Return-State Distribution

Classify each day into one of three stock-return states:

State	Condition on `stock_return`
Loss	`stock_return < 0`
Flat or small gain	`0 <= stock_return < 0.01`
Strong gain	`stock_return >= 0.01`

1. Create a new column called `stock_state`.
2. Count the number of days in each state.
3. Compute the relative frequency of each state.
4. Present the result as a probability distribution table.
5. Check that the estimated probabilities sum to 1.
6. Explain why this distribution is a model if it is used to describe a future trading day.

Exercise 23 - Python: Expected Payoff from an Empirical Distribution

Define a stylized payoff:

Condition	Payoff
Large stock loss day	-3.0%
Otherwise	0.4%

1. Create a column called `strategy_payoff`.
2. Compute the empirical expected payoff as the sample mean of `strategy_payoff`.
3. Compute the empirical standard deviation of `strategy_payoff`.
4. Compute the expected payoff separately on market-stress days and non-stress days.
5. Explain why the expected payoff is not a guarantee of the next day's payoff.

Exercise 24 - Python: Careful Probability Memo

Write a short memo based on the previous Python exercises.

Your memo should mention:

1. the sample size and sample period;
2. how market stress and large stock loss were defined;
3. $P(A)$, $P(B)$, $P(A \cap B)$, and $P(A \cup B)$;
4. $P(B \mid A)$ and $P(B \mid A^c)$;
5. whether the independence benchmark looks plausible in this sample;
6. the empirical expected payoff from Exercise 23;
7. the limitation that these are sample estimates based on historical data, not guarantees about the future.